

Wavelet Theory

$\psi(t)$: Wavelet

i) $\int_{-\infty}^{\infty} \psi(t) dt = 0$ 'oscillatory'

ii) $\int_{-\infty}^{\infty} |\psi(t)|^2 dt < \infty$ finite energy.

$$\psi_{a,b}(t) = \frac{1}{\sqrt{a}} \psi\left(\frac{t-b}{a}\right)$$

Cont. Wavelet Transform & its Inverse

$$f(t) \longleftrightarrow W(a, b) \quad a = \text{scale}, b = \text{shift}$$

$$W(a, b) = \int_{-\infty}^{\infty} f(t) \frac{1}{\sqrt{a}} \psi^*\left(\frac{t-b}{a}\right) dt \quad \begin{matrix} 0 < a < \infty \\ -\infty < b < \infty \end{matrix}$$

$$f(t) = \frac{1}{c} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} W(a, b) \psi_{a,b}(t) \frac{da db}{|a|^2}$$

$$c \equiv \int_{-\infty}^{\infty} \frac{|\psi(\tau)|^2}{|\tau|^2} d\tau \quad 0 < c < \infty.$$

Dyadic Wavelet Decomposition

let us sample 'a' at $a = 2^{-j}$ $-\infty < j < \infty$
and 'b' at $b = 2^j k$

Then $\psi_{j,k}(t) = 2^{j/2} \psi(2^j t - k)$

Now $W(a, b)$ at $a = 2^{-j}$
 $b = k 2^{-j}$

$W(2^{-j}, k 2^{-j})$ will be denoted as
 "d_{JK}"

i.e. $d_{JK} = \text{Coef of } 2^{j/2} \psi(2^j t - k)$

Dyadic Wavelet expansion (decomposition)
 of a function $f(t)$ is now given by

$$f(t) = \sum_J \sum_K d_{JK} \psi_{JK}(t)$$

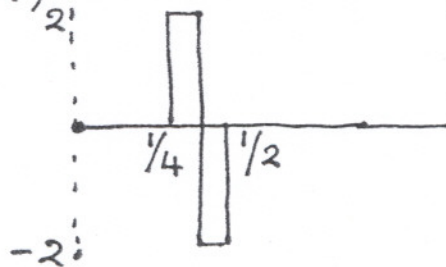
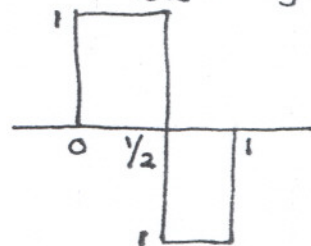
If ψ_{JK} are orthogonal
 $d_{JK} = \langle f(t), \psi_{JK}^* \rangle$

Haar Wavelet

$$\psi(t) = \psi_{1,0}(t)$$

$$\psi_{2,1}(t) = 2^{1/2} \psi(2t - 1)$$

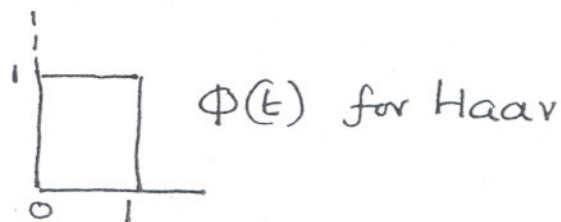
$$\text{Energy} = 2^2 \cdot \frac{1}{4} = 1$$



Haar Scaling function

$$\phi(t)$$

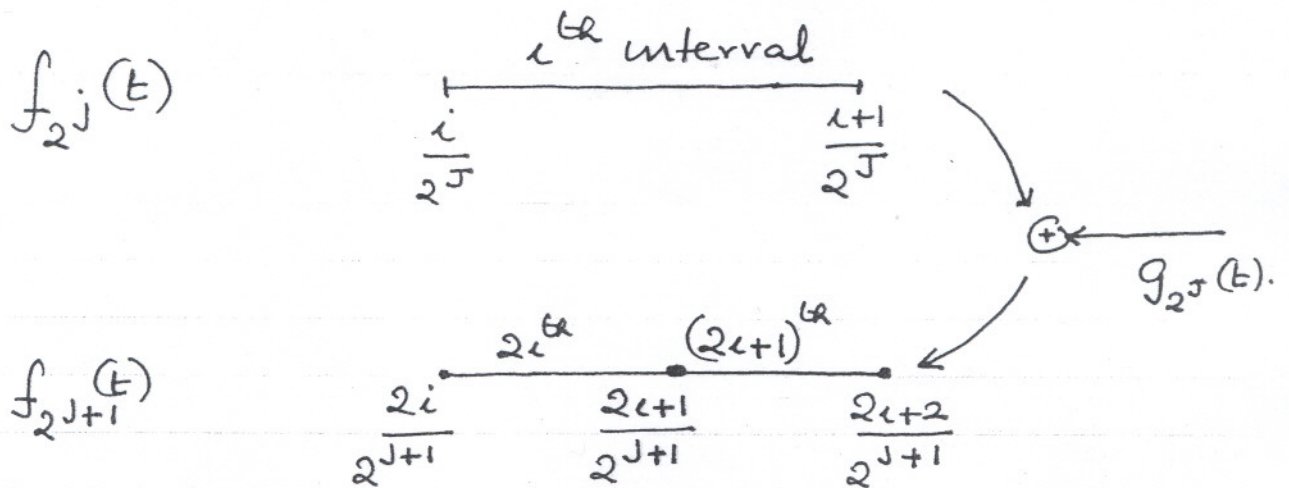
Haar Scaling function



Linear Piecewise Constant approx of a signal $f(t)$

Corresponds

This kind of approx ~~leads~~ to Haar Wavelet/
Scaling function application to $f(t)$

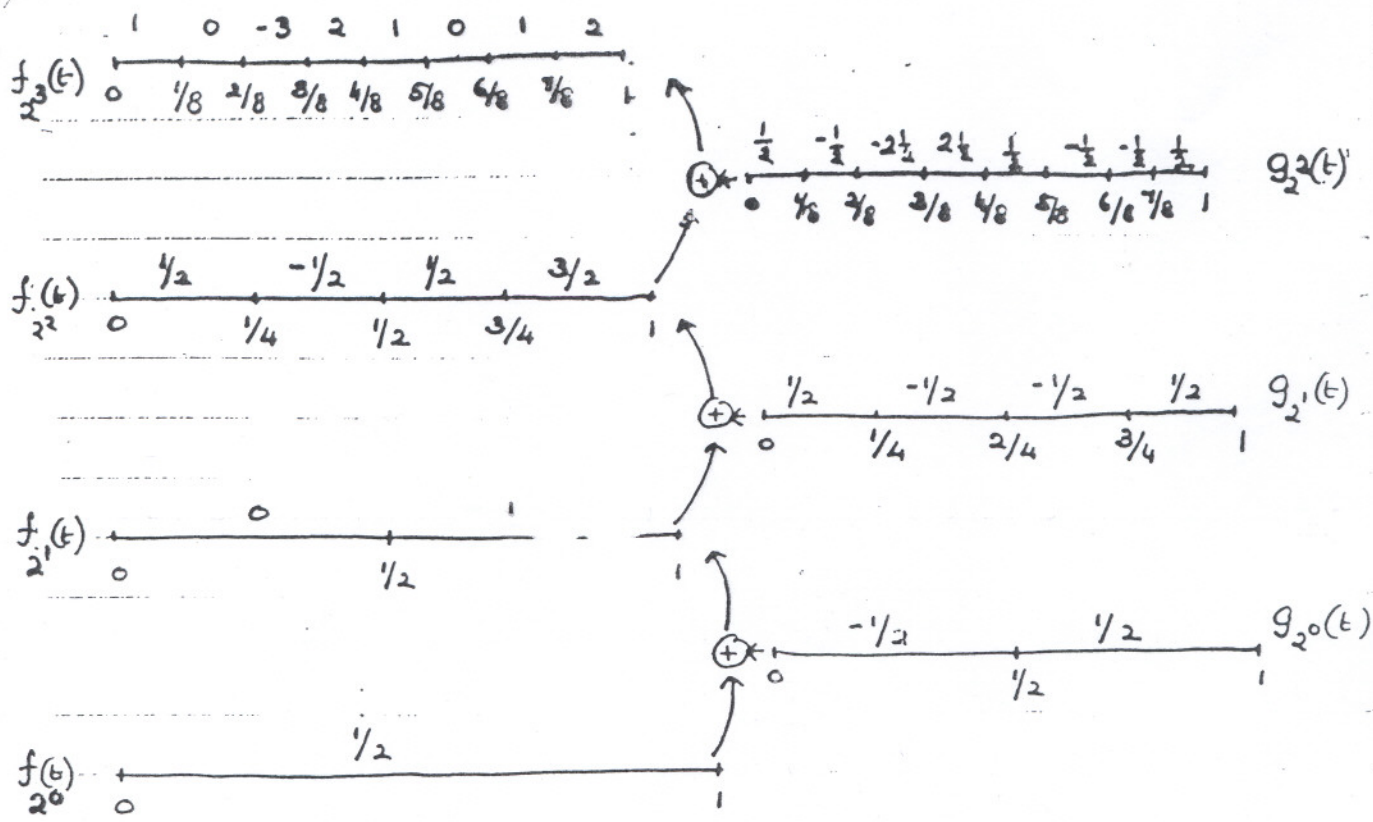


$g_{2^J}(t)$ = detail signal at level J

$f_{2^J}(t)$ = approx at level J of the signal $f(t)$

$$g_{2^J}(t) = f_{2^{J+1}}(t) - f_{2^J}(t)$$

$$\lim_{J \rightarrow \infty} f_{2^J}(t) = f(t) = \sum_{J=-\infty}^{\infty} g_{2^J}(t)$$



$f_{2^J}(t)$ Can be expressed as linear combination of ψ_{JK}

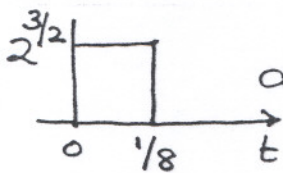
$g_{2^J}(t)$ " " " Φ_{JK}

Eg: $g_{2^3}(t)$ Can be expressed as a l.c of $\psi_{30}(t)$ and its translates $\psi_{31}, \psi_{32}, \psi_{33}$

$g_{2^2}(t)$ $\rightarrow$ Can be expressed as l.c of $\psi_{20}(t)$ and its translates $\psi_{21}, \psi_{22}, \psi_{23}$

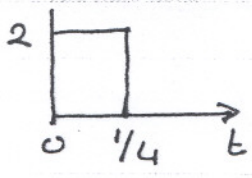
Similarly

$f_{23}(t)$ can be expressed as a l.c of

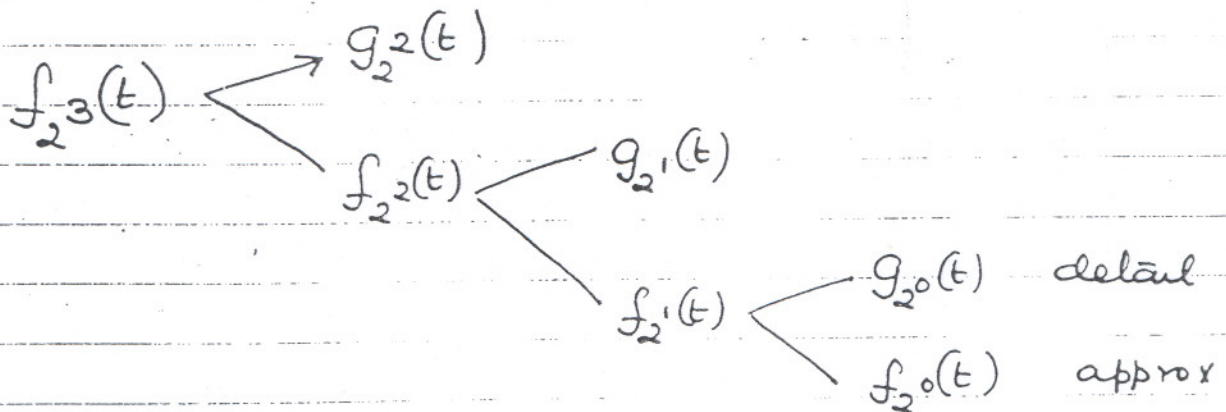
$\Phi_{30}(t)$  and its translates $\Phi_{31}, \Phi_{32}, \dots$

$$\Phi_{30}(t) = 2^{3/2} \phi(2^3 t)$$

$f_{22}(t)$ can be expressed as a l.c of

$\Phi_{20}(t)$  and its translates $\Phi_{21}, \Phi_{22}, \Phi_{23}, \dots$

Decomposition



$$f_{23}(t) = g_{22}(t) + g_{21}(t) + g_{20}(t) + f_{20}(t)$$

$$f_2(t) = [d_{20}\psi_{20} + d_{21}\psi_{21} + d_{22}\psi_{22} + d_{23}\psi_{23}] \\ + [d_{10}\psi_{10} + d_{11}\psi_{11}] + [d_{00}\psi_{00}] + c_{00}\phi_{00}$$

$$[d_{20}, d_{21}, d_{22}, d_{23}] = \left[\frac{1}{4}, -\frac{5}{4}, \frac{1}{4}, -\frac{1}{4} \right]$$

$$[d_{10}, d_{11}] = \left[+\frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}} \right]$$

$$d_{00} = -\frac{1}{2} \quad c_{00} = \frac{1}{2}$$

$$f_2 \rightarrow \left[\frac{1}{2\sqrt{2}}, 0, -\frac{3}{2\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}}, 0, \frac{1}{2\sqrt{2}}, \frac{1}{\sqrt{2}} \right] \begin{bmatrix} \left[\frac{1}{4}, -\frac{5}{4}, \frac{1}{4}, -\frac{1}{4} \right] \{\psi_{2k}\} \\ \left[\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{4} \right] \{\psi_{1k}\} \\ \left[\frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}} \right] \{\psi_{0k}\} \\ \left[0, \frac{1}{\sqrt{2}} \right] \{\psi_{0k}\} \\ \left[-\frac{1}{2} \right] \{\psi_{0k}\} \\ \left[-\frac{1}{2} \right] \{\psi_{0k}\} \end{bmatrix}$$

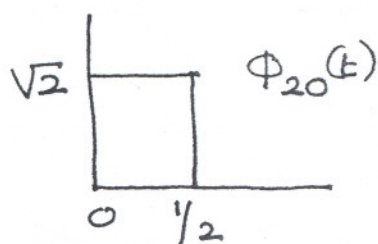
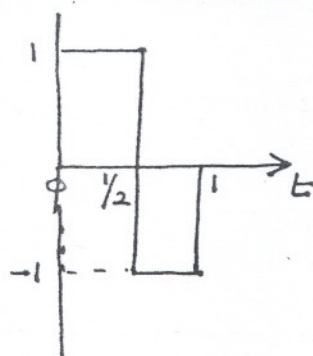
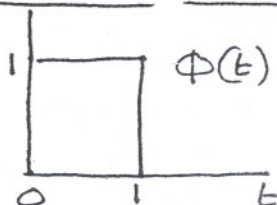
$$\phi_{JK} = 2^{J/2} \phi(2t-k)$$

$$\psi_{JK} = 2^{J/2} \psi(2t-k)$$

$$[c_{30}, c_{31}, c_{32}, c_{33}, c_{34}, c_{35}, c_{36}, c_{37}, c_{38}] \begin{bmatrix} G \\ H \\ H \\ H \end{bmatrix} \begin{bmatrix} [d_{20}, d_{21}, d_{22}, d_{23}] \\ [d_{10}, d_{11}] \\ [d_{00}] \\ [c_{00}] \end{bmatrix}$$

$$\phi(t) = \sum R_k(t) \sqrt{2} \phi(2t-k)$$

Relation between Haar Wavelet & Scaling function



$$\Phi(t) = \Phi_{00}(t) = \frac{1}{\sqrt{2}} \Phi_{20}(t) + \frac{1}{\sqrt{2}} \Phi_{21}(t)$$

(A) $\Phi(t) = \sum_k h(k) \Phi_{2k}(t)$ (in general) $\longleftrightarrow h(0) = h(1) = 1/\sqrt{2}$

$$\psi(t) = \frac{1}{\sqrt{2}} \Phi_{20}(t) - \frac{1}{\sqrt{2}} \Phi_{21}(t)$$

(B) $\psi(t) = \sum_k g(k) \Phi_{2k}(t)$ (in general) here $k=0, 1$

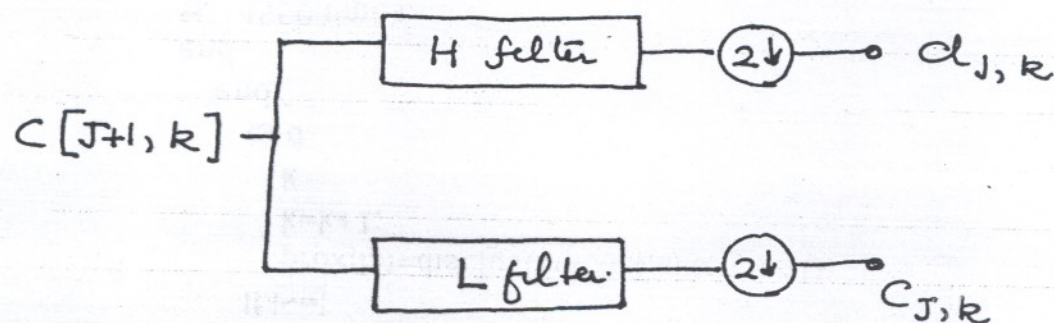
One can show $g(k) = (-1)^k h(-k+1)$

eg: $g(0) = h(1) = 1/\sqrt{2}$

$g(1) = -1 \times h(0) = -1/\sqrt{2}$

A & B are called Dilation or two scale equation

$$C_{(J+1,k)} \begin{cases} d_{Jk} & k = \dots, 0, 1, 2 \\ c_{Jk} & k = -1, 0, 1, \dots \end{cases}$$



$$\text{H filter} : h_{\psi}(-n)$$

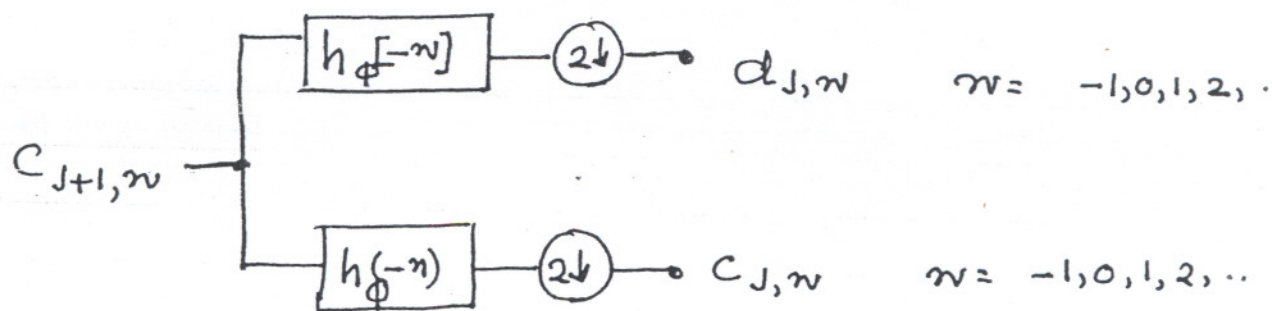
$$h_{\phi}[n] = \{h_0, h_1\}$$

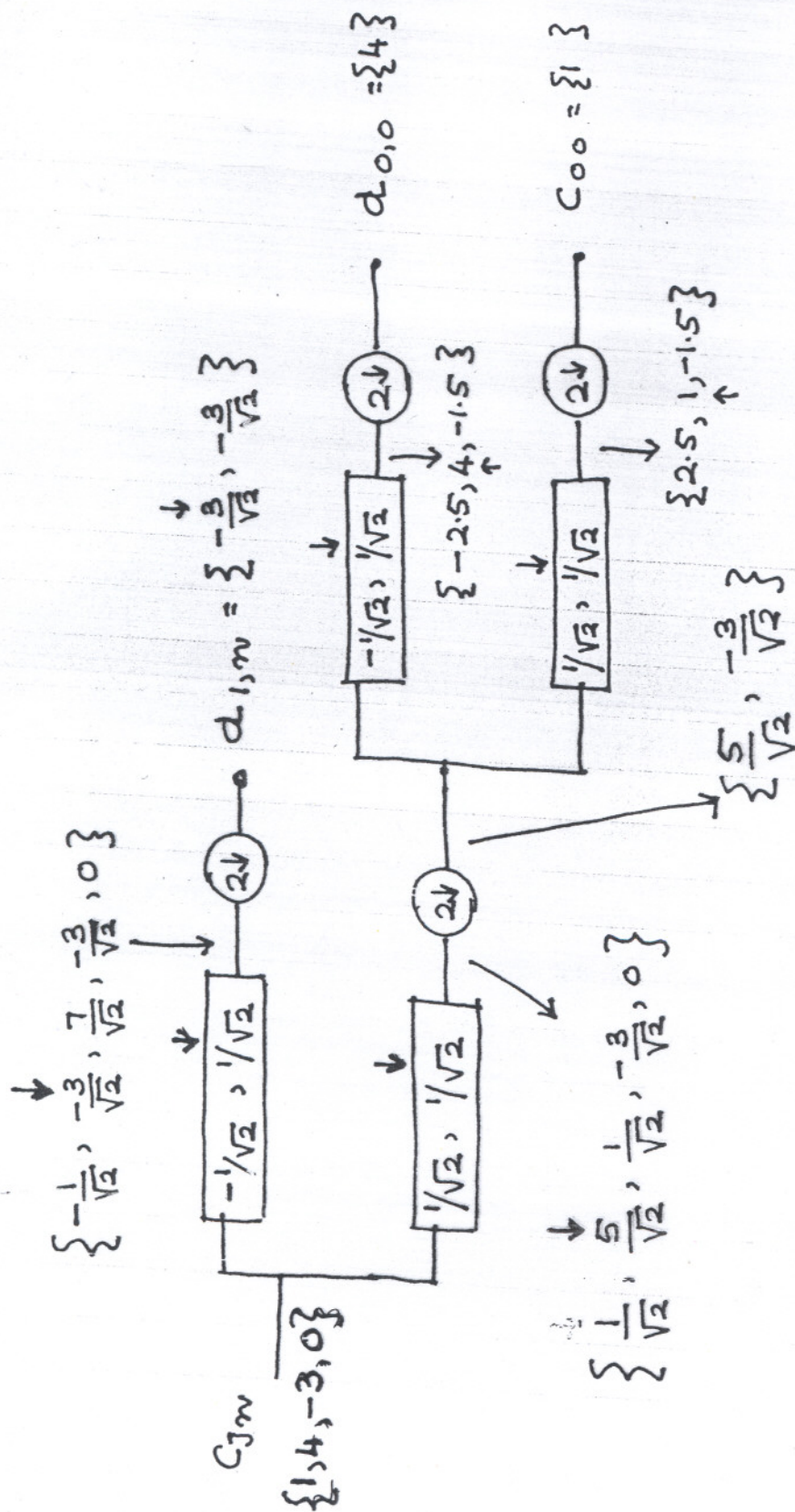
$$\text{L filter} : h_{\phi}(-n)$$

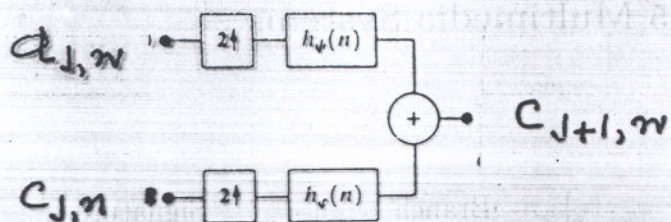
$$h_{\psi}[n] = \{g_0, g_1\}$$

$$d(J,k) = \sum_m h_{\psi}(m-2k) C(J+1,k)$$

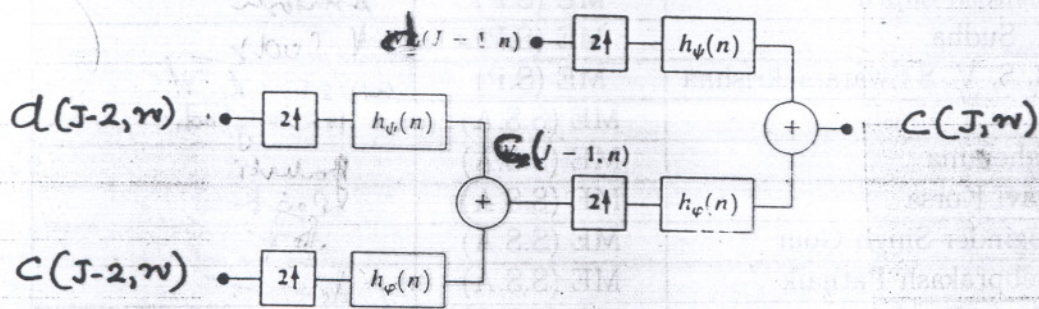
$$c(J,k) = \sum_m h_{\phi}(m-2k) C(J+1,k)$$



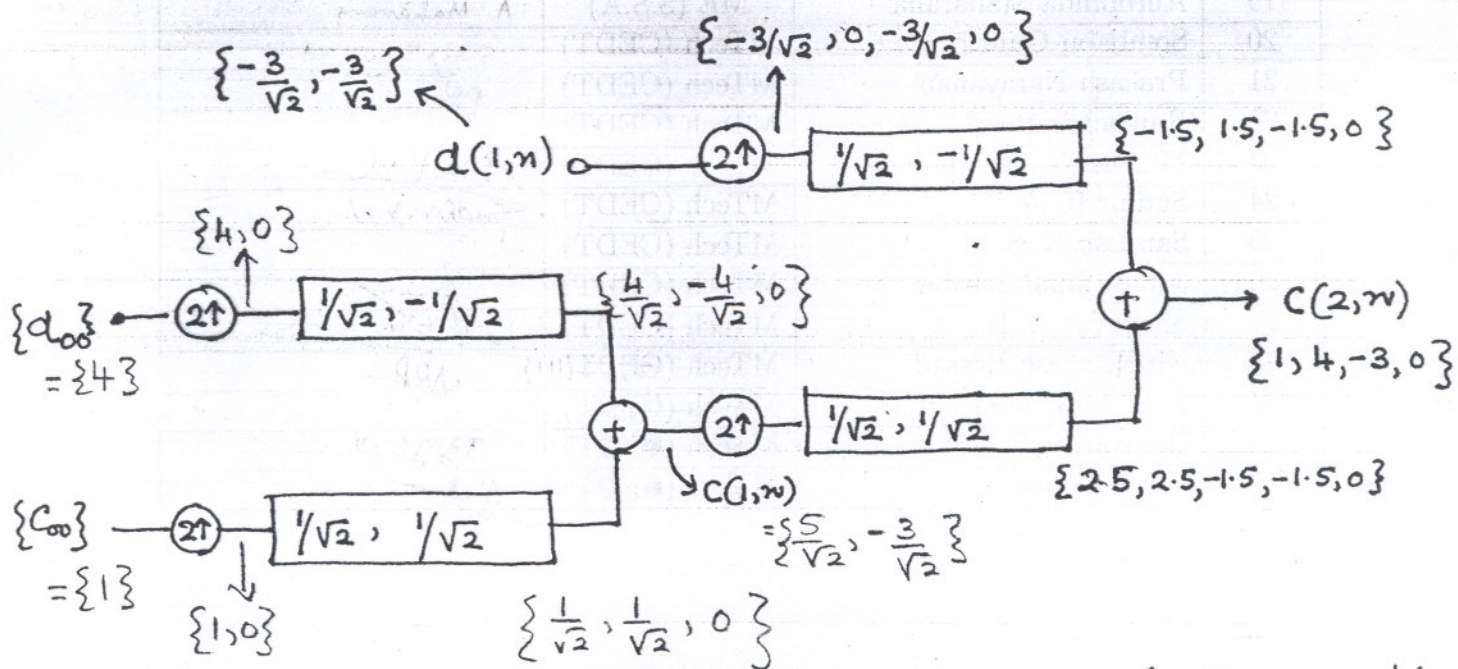




Synthesis filter.



2-stage synthesis filter.



2-scale Inverse Wavelet Transform.

FIGURE 7.15 An FWT analysis bank

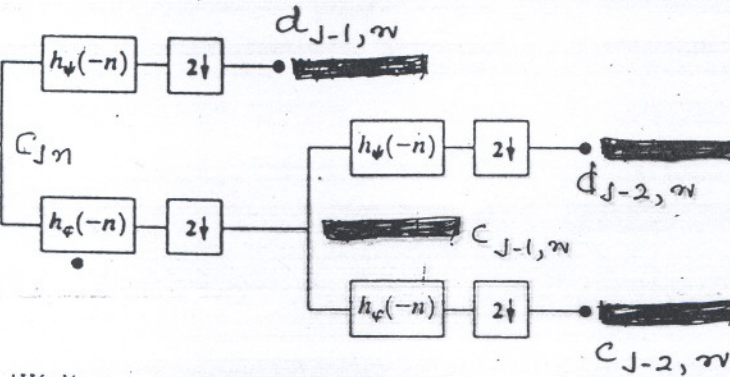
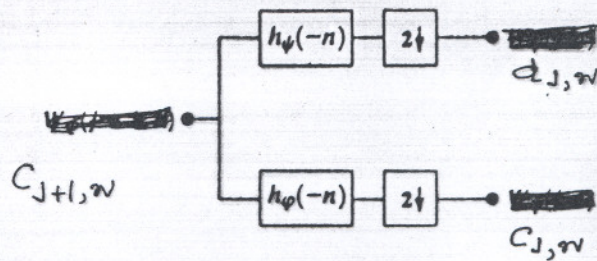


FIGURE 7.16

(a) A two-stage or two-scale FWT analysis bank and (b) its frequency splitting characteristics.

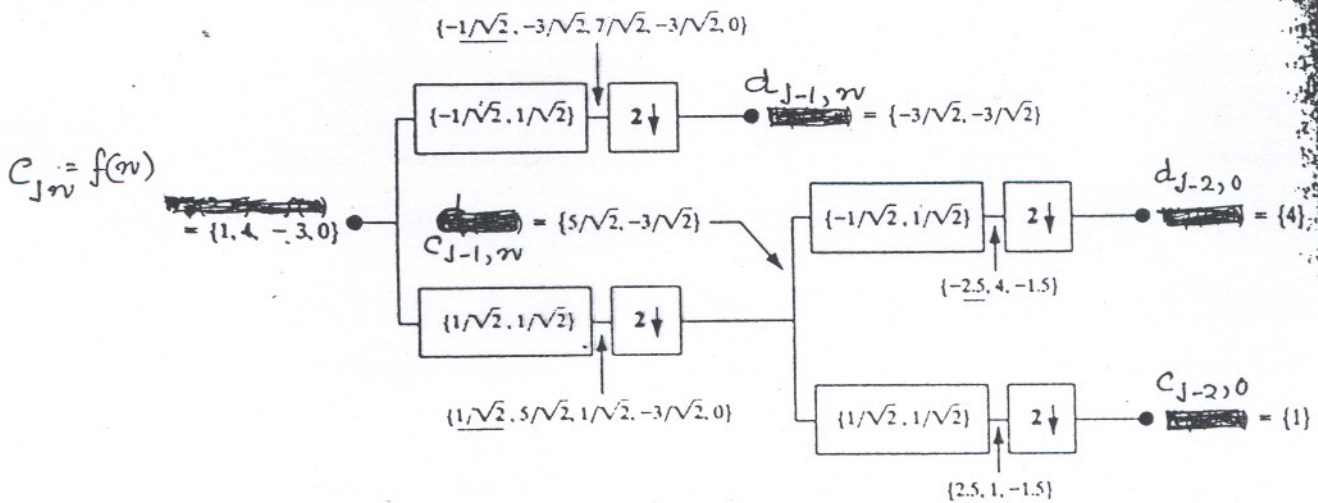
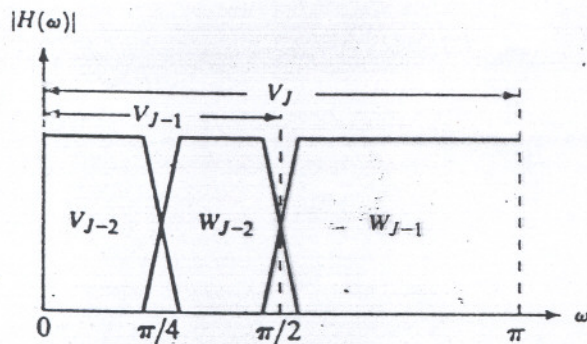


FIGURE 7.17 Computing a two-scale fast wavelet transform of sequence $\{1, 4, -3, 0\}$ using Haar scaling and wavelet vectors.

$$\downarrow$$

$$-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$$

$$0 \quad -3 \quad 4 \quad 1$$

$$\frac{1}{\sqrt{2}}, \frac{3}{\sqrt{2}}$$

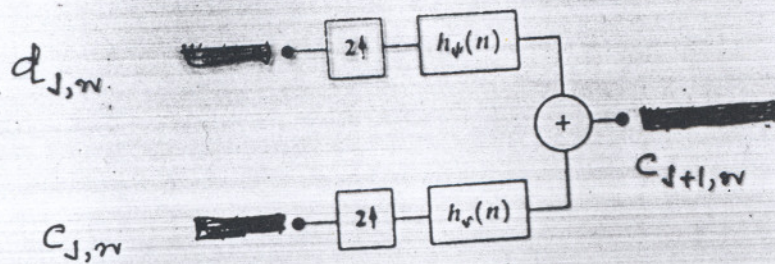


FIGURE 7.18 The FWT^{-1} synthesis filter bank.

FIGURE 7.19 A two-stage or two-scale FWT^{-1} synthesis bank.

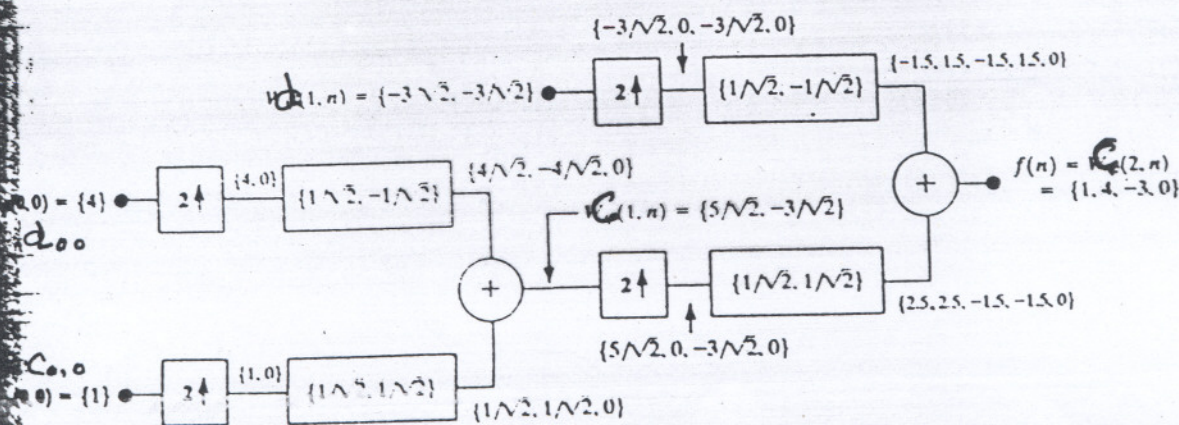
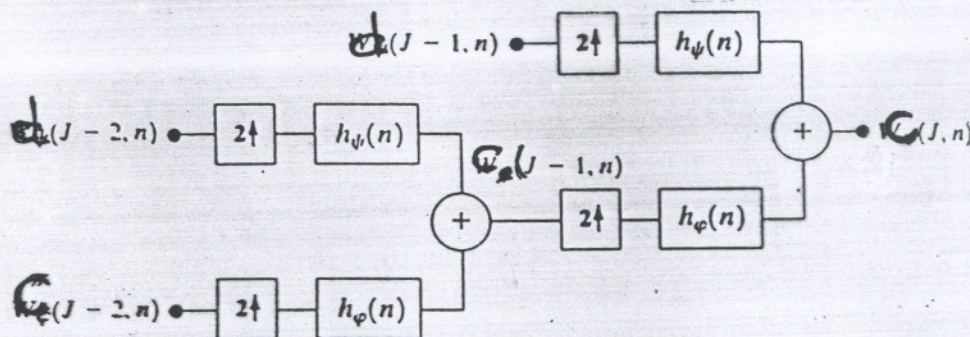


FIGURE 7.20 Computing a two-scale inverse fast wavelet transform of sequence $\{1, 4, -1.5\sqrt{2}, -1.5\sqrt{2}\}$ with scaling and wavelet vectors.